

Cheng Dai

[[Email](#)] | [[Github](#)] | [[Homepage](#)] | [[Google Scholar](#)] | [Beijing, China]

OVERVIEW

I'm a researcher specializing in thermal sensing, video generation, and physical AI.

My research investigates how generative models can learn the dynamics, geometry, and physical structure of real-world environments, and how these models can support perception, planning, and interaction.

POSITIONS

Tsinghua University, Institute for AI Industry Research

Jul. 2026 - ?

Research Assistant

Supervisors: [Chaojian Li](#) and [Hao Zhao](#)

- Dissertation Fields: Quantum Machine Learning, and World Model.
- Collaborating with technical teams from BAAI, HKUST and QBoson Inc.

Westlake University, School of Science

Sep. 2025 - Jul. 2026

Research Assistant

Supervisors: [Xin Yuan](#) and [Fanglin Bao](#)

- Dissertation Fields: HADAR (nature2023, Temperature-emissivity-texture-distance decomposition)
- AI department manager in the flab. Conduct Research and guide students.
- Release the first thermal hyperspectral image restoration methods based on Radiative Transfer Equation, **HAIR**.
- Release the first temperature-emissivity-texture decomposition dataset and benchmark, **TeX-1500**.
- Release the first sparse-active thermal texture imaging framework that isolates reflection-dominant thermal texture through controlled LWIR illumination and leverages RGB-domain pretrained representations for reconstruction, **T²exture**.

EDUCATION

Wuhan University

Sep. 2025 - Mar. 2026

Graduate Study in State Key Laboratory of Surveying and Mapping Remote Sensing Information Engineering (LIESMARS), Major in Communication and Information Systems

Supervisors: [Wei He](#)

- GPA: 90.79% (among 6 postgraduate courses). recipient of the First-Class Graduate Scholarship.
- Voluntarily withdrew after consultation with my supervisor to pursue a more suitable research direction in physical AI.

Jilin University

Sep. 2021 - Jun. 2025

BSc. Major in Communication Engineering

- GPA: 90.58%. Rank: 12/283 (Outstanding graduate, top 5%), 508/710 in CET6. (German and Russian study for 2 years)
- Outstanding Undergraduate Thesis (top 3%).
- Outstanding Student of the School (*4). First-Class Scholarship, Second-Class Scholarship (*3)

PUBLICATIONS

[1] **Cheng Dai***, Jiale Lin*, Hongyi Xu, Bingxuan Song, Ziyang Xie, Fanglin Bao

TeX-1500: A Paired Real-World LWIR Hyperspectral Dataset and Benchmark for Temperature-Emissivity-Texture Decomposition. [[Paper](#)], [[Homepage](#)], [[huggingface](#)] (arxiv version, published in the future)

[2] **Cheng Dai***, Jiale Lin*, Bingxuan Song, Yifei Chen, Jiashuo Chen, Xin Yuan, and Fanglin Bao

HAIR: HADAR-Based Thermal Infrared Hyperspectral Image Restoration. [[Paper](#)], [[Homepage](#)] (IEEE **TIP** under review)

[3] Jiashuo Chen*, **Cheng Dai*** (**co-first author and listed alphabetically**), Yanan Hu and Fanglin Bao

T2exture: Sparse Active Thermal Texture Imaging. [[Paper](#)], [[Homepage](#)], [[huggingface](#)] (**AAAI 2027** under review)

[4] Hongyi Xu*, Jiale Lin*, **Cheng Dai**, ..., Liqin Cao, Yanfei Zhong, Yiyuan She, Fanglin Bao.

Universal computational thermal imaging overcoming the ghosting effect. [[Paper](#)] (**Optica** Under review)

INTERNSHIP

Lightwheel, Embodied Assets Group

Dec. 2024 - Feb. 2025

Group Leader: [Jian Yang](#), and [Soroush Nasiriany](#)

- **Authored and validated the first 20 high-fidelity, interactive household [manipulation scenes](#)** for [RoboCasa365](#), spanning asset construction, physical configuration and validation; established reusable templates for subsequent scaling.
- Built a SpaceMouse teleoperation interface and bidirectional Blender ↔ YAML tooling, enabling rapid scene authoring and [validation](#) across Blender and RoboCasa/MuJoCo; **reduced validation iteration time by 20 ×**.
- Contributed to the embodied simulation, asset-generation, data-collection, and teleoperation stack using Blender, MuJoCo, RoboSuite, Isaac Sim, and RoboCasa; upstreamed contributions to [RoboCasa](#).

Amoon AI, Large Audio Model Department

Apr. 2025 - Sep. 2025

- Independently collect and label 20k infant-cry audios (update 6 versions), and train or finetune models to classify the baby cries and detect activities. Involved Models: BEATs, Audio-MAE, Wav2vec2, Qwen-audio, Whisper, Conformer.
- Achievements: reach 67.2% acc in five-class (diaper, hunger, hug, sleepy, gasping) edge deployment device.

AWARDS IN COMPETITION

- **First Prize** in the Provincial Level of the National Undergraduate Mathematics Competition.
- Honorable Mention in the American Mathematical Competition.
- Student Leader in Science and Technology Innovation Plan (Changchun Institute of Optics, CAS), **excellent completion**.
- Student Leader in Provincial-level college student innovation and entrepreneurship competition, successful completion.

SKILLS

- **Generative AI & Foundation Models:** Pre-training and fine-tuning of LLMs, VLMs, and audio generative models; diffusion-model improvement, multi-branch diffusion, Schrödinger Bridge, and inference acceleration.
- **Machine Learning Theory:** Model interpretability and theoretical analysis using Fisher information, convex optimization, and differential-equation-based modeling.
- **Multimodal, Video & Graphics:** Flow-based video interpolation, video generation, and rendering/graphics systems.
- **Scientific Inverse Problems:** Model-based physical inversion, numerical methods, and hyperspectral image processing.
- **Time-Series & Signal Processing:** PatchTST, DLinear, TimesFM; EEG, ECG, and geophysical signal analysis.
- **Quantum Computing:** Quantum computing and quantum machine learning.

Software: [Cuda-Jax, Latex, Codex, Git, Ubuntu, Python, C++, FFmpeg, Html5, CSS3, Javascript, MATLAB].